Berlin Analytics & AI Hackathon

Guide for the Hackathon, March 23rd 2026

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What is Snowflake?

5-Minute Crash Course

Snowflake: Enterprise Cloud Data Platform

Snowflake is a unified cloud data platform that delivers:

- **Data Storage**: Centralized data warehousing and management
- **Analytics**: Advanced query processing and business intelligence
- **AI/ML Development**: Native AI and machine learning capabilities
- **Web-Based Access**: Intuitive browser interface via Snowsight

📺 **Watch This First:**

Snowflake: The Platform For The AI Era - *1:12 minutes*

What is Snowflake? Developer Demo (2026) - *11:12 minutes*

KEY TERMS MADE SIMPLE		
TERM	WHAT TO DO	WHY IT MATTERS
Database	Container for your data	Organizes your data by project/domain
Schema	Logical grouping within database	Further organizes tables by function
Table	Where actual data is stored	Like a spreadsheet with rows & columns
Warehouse	Computing power to process data	Right-sizing optimizes cost-to-performance ratio for your workload (performance scales with size: XS-6XL)
Query	Question you ask your data	Returns insights from your data
Cortex Code	AI assistant that helps you explore data	Writes code for you in plain English

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Finding Your Challenge Data:

1. Click **"Data"** in left menu
2. Look for **"Databases"**
3. Navigate to your challenge database/table
4. Click to preview your data

First Time Running a Query?

1. Click **"Projects"** → **"Worksheets"**
2. Click **"+ Worksheet"**
3. Type: `SELECT * FROM [YOUR_TABLE] LIMIT 10;`
4. Click **"Run"** (or Ctrl+Enter)

But you don't need to write SQL! Cortex Code does it for you in plain English.

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Getting Started

Step 1: Log into Snowsight

- Go to your Snowflake URL: `https://[your-account].snowflakecomputing.com`
- Enter your credentials
- You're in! ✅

Step 2: Find Cortex Code

Choose your preferred method:

Option A: Snowsight (Recommended - No Installation)

- Look for the Cortex Code icon in **bottom right corner**
- OR go to **Projects** → **Workspaces** → **Cortex Code**
- Works immediately in your browser

Option B: Cortex Code CLI (Advanced Users)

- More powerful for developers
- Requires installation
- Works from your terminal/command line

Step 3: Start Exploring!

- Type your question in plain English
- Press Enter
- Watch Cortex Code help you explore your data!

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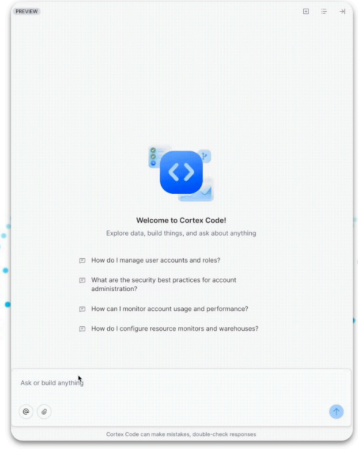
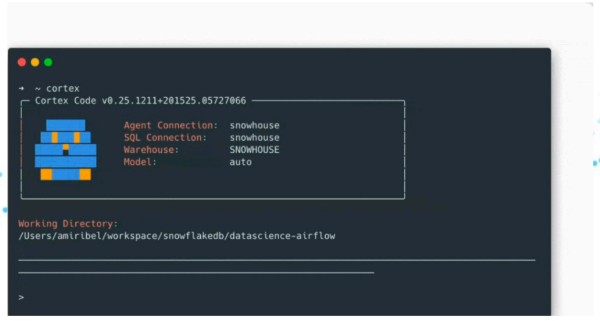
Cortex Code: CLI vs Snowsight

How can you access Cortex Code?





Snowsight **CLI**

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The Difference is Access & Orchestration

	CORTEX CODE IN SNOWSIGHT	CORTEX CODE CLI
Environment	Web based, inside the Snowsight UI	Your terminal/ IDE; supports macOS/Linux, and Windows
Primary Access	Snowflake metadata, workspaces, account info	Local filesystem, code repos, git, dbt, and developer tools
Best For	• SQL Development • Admin Tasks • Data Discovery • Data science and ML in notebooks	• End-to-End Projects • Agents & App Deployment • Pipeline Orchestration • Agent Skill Development
Example Task	“Refine this SQL query for better performance”	"Create a new mart model in my dbt project and write tests for it"

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Cortex Code - Your AI Assistant

Cortex Code is an AI-powered assistant that helps you across the entire data lifecycle:

💬 HOW IT WORKS

Just describe what you want in plain English - Cortex Code generates the code and executes it for you.

🔧 DATA ENGINEERING

Build pipelines, transform data, ensure quality

→ See prompts here

📊 ANALYTICS

Create visualizations, calculate metrics, find insights

→ See prompts here

🤖 DATA SCIENCE & AI

Build ML models, create agents, make predictions

→ See prompts here

🆘 GET HELP

Questions or stuck? Quick help available

→ See troubleshooting here

🎮 Prompts to Try

Just type these **prompts** in the Cortex Code chat box.

DATA ENGINEERING

Use Cortex Code to build and manage your data infrastructure:

📥 LOAD & TRANSFORM DATA

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```
None
Load the CSV file from my stage into the customer_data table
```

```
None
Create a data pipeline that extracts data from [SOURCE], transforms it by
[RULES], and loads it into [TARGET]
```

```
None
Join the sales_data and customer_data tables on customer_id
```

✨ DATA QUALITY

```
None
Generate a data quality report showing completeness, uniqueness, and validity
metrics
```

```
None
Check this dataset for missing values, duplicates, and outliers
```

```
None
Create alerts for when data quality drops below 95%
```

⚡ PERFORMANCE OPTIMIZATION

```
None
Analyze query performance and suggest optimizations
```

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```
None
Recommend optimal warehouse size for this workload
```

⏰ AUTOMATION

```
None
Schedule a task to run this data refresh daily at 2 AM
```

ANALYTICS

Use Cortex Code to analyze data and create insights:

📈 EXPLORE YOUR DATA

```
None
Summarize the data in [YOUR_DATABASE].[YOUR_SCHEMA].[YOUR_TABLE]
```

```
None
What interesting patterns can you find in this dataset?
```

```
None
Show me the top 5 most interesting insights from this data
```

📊 VISUALIZATIONS & DASHBOARDS

```
None
Create a chart showing [something from your data]
```

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```
None
Build a simple dashboard with 3 key metrics from this dataset
```

```
None
Show me trends over time in this data
```

💼 BUSINESS METRICS

```
None
Calculate key business metrics: CAC, LTV, churn rate, MRR
```

```
None
What trends do you see in this data?
```

```
None
If I wanted to present this data to executives, what should I highlight?
```

🎯 ADVANCED ANALYSIS

```
None
Segment customers into groups based on purchasing behavior
```

```
None
Analyze which marketing channels drive the most conversions
```

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```
None
Find correlations I probably wouldn't notice on my own
```

DATA SCIENCE & AI

Use Cortex Code to build ML models and AI agents:

🤖 BUILD AI AGENTS

```
None
Build a simple agent that can answer questions about this data
```

```
None
Create a chatbot that knows about this dataset
```

📉 PREDICTIVE ANALYTICS

```
None
Build a forecast model for next quarter's sales based on historical trends
```

```
None
Predict customer churn probability based on behavior patterns
```

```
None
Create a recommendation engine for product suggestions
```

🧠 MACHINE LEARNING

```
None
Build a classification model to predict [target variable]
```

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```
None
Train a model to identify anomalies in this dataset
```

```
None
Create features for ML: customer lifetime value, recency, frequency
```

🔍 EXPLORATORY

```
None
What's the most surprising thing about this data?
```

```
None
If this data could tell a story, what would it be?
```

```
None

Suggest potential use cases for ML on this dataset
```

🆘 QUICK HELP

```
None
What can you help me do with this data?
```

```
None
I'm new to this data - what should I explore first?
```

```
None
What does this error mean?
```

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```
None
Help me understand what [COLUMN_NAME] represents
```

```
None
Explain this to me in simpler terms
```

```
None
I'm stuck, what should I try?
```

Resources & Learning

⭐ Must Have (Start Here!)

- **Cortex Code Workflow Examples**: https://docs.snowflake.com/en/user-guide/cortex-code/workflows
- **Build an Agent from Scratch**: https://www.snowflake.com/en/developers/guides/build-a-cortex-agent-from-scratch-with-snowflake/
- **Quickstarts Hub**: https://quickstarts.snowflake.com → Interactive tutorials with sample data

🎓 Snowflake Basics

- **Snowflake User Guide**: https://docs.snowflake.com/en/user-guide
- **Sample Templates**: http://app.snowflake.com/templates
- **Snowflake for Developers:** https://www.snowflake.com/en/developers/
- **What is a Virtual Warehouse?:** Search "warehouses" in docs
- **Database Objects Explained:** Search "databases tables views" in docs

🔧 Data Engineering

- **Data Loading**: https://docs.snowflake.com/en/user-guide/data-load-overview
- **Transformations**: https://docs.snowflake.com/en/user-guide/tables-storage-considerations

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- **Task Scheduling**: https://docs.snowflake.com/en/user-guide/tasks-intro

📊 Analytics

- **Window Functions**: https://docs.snowflake.com/en/sql-reference/functions-analytic
- **Time-Series Analysis**: https://docs.snowflake.com/en/user-guide/querying-time-series
- **Statistical Functions**: https://docs.snowflake.com/en/sql-reference/functions-aggrega

🧠 Data Science & ML

- **Cortex ML Functions:** https://docs.snowflake.com/en/user-guide/snowflake-cortex/ml-functions
- **Build ML Models:** https://docs.snowflake.com/en/user-guide/ml-powered-features
- **Cortex AI Overview:** https://docs.snowflake.com/en/user-guide/snowflake-cortex/overview

🤖 Cortex Code & Agents Tutorial

- **CLI Reference**: https://docs.snowflake.com/en/user-guide/cortex-code/cli-reference
- **Agent Tutorials**: https://docs.snowflake.com/en/user-guide/snowflake-cortex/cortex-agents-tutorials
- **Manage Agents**: https://docs.snowflake.com/en/user-guide/snowflake-cortex/cortex-agents-manage

📺 YouTube Channels

- https://www.youtube.com/@snowflakedevelopers
- https://www.youtube.com/@Snowflake

🎯 Snowflake Quickstarts:

From Zero to Snowflake (20 - 30 mins):

- **Link**: https://www.snowflake.com/en/developers/guides/zero-to-snowflake/
- **Why**: Learn by doing with real examples

Getting Started Guides (15 mins):

- **Link**: https://docs.snowflake.com/en/user-guide-getting-started
- **Why**: Understand key concepts

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Snowsight Quick Tour (10 mins):

- **Link**: https://docs.snowflake.com/en/user-guide/ui-snowsight-quick-tour
- **Time**: 10 minutes
- **Why**: Learn where everything is

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